

BUSINESS INTELLIGENCE PORTFOLIO PROJECT

Healthcare Operations & Patient Performance Analytics

End-to-End Data Preparation, Multi-Fact BI Modeling,
KPI Analysis & Executive Dashboarding

Technology Stack

Python • Google Colab • Pandas • Excel • Tableau Public

Project Scope

Synthetic Saudi Healthcare Network
2023-2025 Analytical Period

PROJECT DOCUMENTATION

Document Control

Item	Details
Project Name	Healthcare Operations & Patient Performance Analytics
Document Type	Project Documentation / BI Case Study
Analytical Period	January 2023 – December 2025
Data Environment	Synthetic Saudi healthcare network; SAR currency
Primary BI Platform	Tableau Public
Data Preparation	Python / Pandas in Google Colab
Primary Output	Six interactive Tableau dashboards
Supporting Outputs	Prepared Excel analytical workbook, notebook, Tableau workbook and six dashboard PDFs
Status	Completed

Purpose: This document explains the business problem, data architecture, preparation workflow, analytical model, KPI definitions, dashboard design, findings, recommendations, validation and portfolio deliverables.

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1. Executive Summary

Healthcare Operations & Patient Performance Analytics is an end-to-end Business Intelligence portfolio project designed to demonstrate how a healthcare organization can consolidate operational, financial, patient-experience, utilization, claims, pharmacy, laboratory, imaging and physician-performance data into a decision-ready analytical environment.

The project uses a synthetic Saudi healthcare network covering 2023–2025. The analytical environment contains 75,000 encounters across 30,000 patients, 10 facilities, 15 departments, 60 physicians, 25 services, 30 diagnoses and 8 payers. Supporting fact tables extend analysis into appointments, bed utilization, claims, patient surveys, pharmacy, laboratory transactions and imaging.

Data preparation was performed in Python/Pandas using Google Colab. The workflow standardized fields, derived analytical measures and prepared a relational workbook for Tableau. Tableau Public was then used to construct a multi-fact relationship model and six dashboards covering executive performance, patient/utilization, hospital operations, financial performance, patient experience/quality, and physician/service performance.

Selected control results include SAR 110.1M service revenue, SAR 66.5M treatment cost, SAR 43.6M gross profit, 39.6% gross profit margin, 21.2 minutes average wait time, 4.52/5 average patient satisfaction and a 6.51% readmission rate.

2. Business Context & Problem Statement

Healthcare organizations generate high-volume data across clinical encounters, scheduling, capacity, billing, claims, patient experience and provider activity. When these areas are reviewed independently, management can struggle to see the relationships between demand, operational capacity, financial outcomes and quality indicators.

This project addresses that challenge by creating a unified analytical framework in which shared dimensions filter multiple business processes while preserving the native grain of each fact table. This is important because encounter records, appointments, claims, surveys and facility-day capacity records do not represent the same business event.

Business Need

- Provide a single executive view of healthcare performance.
- Measure patient demand, utilization and appointment efficiency.
- Monitor facility capacity and bed utilization.
- Assess revenue, treatment cost, gross profit and payer collection.
- Track patient satisfaction, outcomes and readmissions.
- Compare physician and service productivity and revenue efficiency.
- Translate analytical results into measurable operational recommendations.

3. Project Objectives

1. Build a reproducible data-preparation workflow using Python/Pandas.
2. Transform raw healthcare data into analysis-ready dimension and fact tables.
3. Preserve fact-table grain and avoid double counting from inappropriate physical joins.

4. Implement a Tableau multi-fact relationship model using shared dimensions.
5. Develop six portfolio-grade dashboards for different management audiences.
6. Create a KPI framework with explicit business targets where appropriate.
7. Validate dashboard measures against source control totals and expected analytical behavior.
8. Produce a reusable portfolio package containing data, notebook, Tableau workbook and dashboard exports.

4. Scope & Analytical Questions

In Scope

- Healthcare encounters and treatment economics.
- Patient demographics and encounter behavior.
- Facility and department operations.
- Appointment volume and cancellation behavior.
- Bed capacity and utilization.
- Claims and payer collection performance.
- Patient satisfaction, outcomes and readmission.
- Physician productivity and service efficiency.
- Pharmacy, laboratory and imaging transactions as supporting domains.

Core Analytical Questions

- How are revenue, cost, profit and encounter volumes trending?
- Which facilities and departments drive activity and revenue?
- Where is operational capacity being utilized most or least?
- What proportion of appointments are cancelled and how does waiting time perform?
- Which payers have stronger or weaker collection performance?
- Is patient satisfaction meeting the 4.5/5 target?
- Which encounter types have elevated readmission rates?
- Which physicians and services generate the strongest revenue and volume?
- Which services produce the strongest revenue per treatment minute?

5. Dataset Overview

The project uses synthetic data designed to resemble a multi-facility Saudi healthcare network. Currency-based measures are represented in Saudi Riyals (SAR). The primary analytical period is 2023–2025.

Table	Business Grain	Rows	Primary Purpose
FACT_ENCOUNTER	One encounter	75,000	Core clinical/financial performance
DIM_PATIENT	One patient	30,000	Patient attributes
DIM_DATE	One date	1,096	Calendar/time analysis
DIM_FACILITY	One facility	10	Facility attributes

DIM_DEPARTMENT	One department	15	Department attributes
DIM_PHYSICIAN	One physician	60	Provider attributes
DIM_SERVICE	One service	25	Service attributes
DIM_DIAGNOSIS	One diagnosis	30	Diagnosis attributes
DIM_PAYER	One payer	8	Payer attributes
APPOINTMENTS	One appointment	5,000	Scheduling and cancellations
BED_UTILIZATION	One facility-day	10,960	Capacity/utilization
CLAIMS	One claim	8,000	Claims and collections
PATIENT_SURVEYS	One survey	6,000	Patient experience
PHARMACY	One pharmacy transaction	7,000	Medication transaction activity
LAB_TRANSACTIONS	One lab test	7,000	Laboratory activity
IMAGING	One imaging transaction	5,000	Imaging activity

6. Data Architecture & Grain

The architecture is intentionally multi-fact rather than a single flattened table. Each fact table represents a different business process and therefore has a different grain. A single physical join across all facts could multiply rows and inflate revenue, cost, appointment counts, claims, survey responses or capacity measures.

Central Fact

FACT_ENCOUNTER is the primary encounter-level fact and contains measures such as service revenue, treatment cost, gross profit, treatment time and total service time. It is related to date, patient, facility, department, physician, service, diagnosis and payer dimensions.

Supporting Facts

APPOINTMENTS, BED_UTILIZATION, CLAIMS, PATIENT_SURVEYS, PHARMACY, LAB_TRANSACTIONS and IMAGING remain separate facts. Shared dimensions are used where their business meaning and keys support cross-process filtering.

Grain Preservation Principle

- FACT_ENCOUNTER = one row per encounter.
- APPOINTMENTS = one row per appointment.
- BED_UTILIZATION = one row per facility-day.
- CLAIMS = one row per claim.
- PATIENT_SURVEYS = one row per survey.
- PHARMACY = one row per pharmacy transaction.
- LAB_TRANSACTIONS = one row per lab transaction.
- IMAGING = one row per imaging transaction.

7. Data Preparation & Quality Process

The data-preparation stage was completed in Google Colab using Python and Pandas. The objective was to create a controlled, reproducible analytical workbook rather than manually prepare data inside Tableau.

1. Load and inspect source tables - review structure, data types, record counts and key fields.
2. Standardized fields - normalize identifiers, dates, categories and numeric measures.
3. Derive analytical measures - create measures needed for revenue, cost, profit, service time, outcomes and KPIs.
4. Validate business logic - check keys, date coverage, cardinalities and control totals.
5. Export analytical workbook - write validated dimension and fact tables to the final Excel workbook.
6. Validate final export - confirm expected row and column counts before Tableau ingestion.

Export Validation Results

- FACT_ENCOUNTER: 75,000 × 33
- DIM_PATIENT: 30,000 × 13
- DIM_DATE: 1,096 × 13
- DIM_FACILITY: 10 × 9
- DIM_DEPARTMENT: 15 × 7
- DIM_PHYSICIAN: 60 × 8
- DIM_SERVICE: 25 × 7
- DIM_DIAGNOSIS: 30 × 4
- DIM_PAYER: 8 × 6
- APPOINTMENTS: 5,000 × 17
- BED_UTILIZATION: 10,960 × 11
- CLAIMS: 8,000 × 17
- PATIENT_SURVEYS: 6,000 × 16
- PHARMACY: 7,000 × 12
- LAB_TRANSACTIONS: 7,000 × 13
- IMAGING: 5,000 × 14

8. Tableau Data Model

Tableau Public was configured using a multi-fact relationship architecture. This allows separate facts to coexist without forcing them into a single physical join and supports analysis at the correct grain.

FACT_ENCOUNTER Relationships

- FACT_ENCOUNTER.Date ID → DIM_DATE.Date ID
- FACT_ENCOUNTER.Patient ID → DIM_PATIENT.Patient ID
- FACT_ENCOUNTER.Facility ID → DIM_FACILITY.Facility ID
- FACT_ENCOUNTER.Department ID → DIM_DEPARTMENT.Department ID
- FACT_ENCOUNTER.Physician ID → DIM_PHYSICIAN.Physician ID
- FACT_ENCOUNTER.Service ID → DIM_SERVICE.Service ID

- FACT_ENCOUNTER.Diagnosis ID → DIM_DIAGNOSIS.Diagnosis ID
- FACT_ENCOUNTER.Payer ID → DIM_PAYER.Payer ID

Supporting Fact Relationships

APPOINTMENTS shares date, patient, facility, department and physician dimensions. BED_UTILIZATION shares date and facility. CLAIMS shares date, patient and payer. PATIENT_SURVEYS shares date and patient. PHARMACY, LAB_TRANSACTIONS and IMAGING share date and patient.

DIM_SERVICE and DIM_DIAGNOSIS remain downstream dimensions associated with FACT_ENCOUNTER and are not used as central shared dimensions across unrelated facts.

Why Relationships Were Used

- Prevent measure multiplication caused by physical joins.
- Preserve each fact's native business grain.
- Allow shared dimensions to filter multiple processes.
- Support cleaner KPI calculations across operational domains.
- Provide a scalable architecture for future fact tables.

9. KPI Framework & Business Targets

The KPI framework combines financial, operational, patient-experience, quality and provider-performance measures. Targets are illustrative management thresholds used to demonstrate performance monitoring.

KPI	Definition / Logic	Observed Result	Target
Service Revenue	SUM(Service Revenue SAR)	SAR 110.1M	-
Gross Profit	Revenue – Treatment Cost	SAR 43.6M	-
Profit Margin	Gross Profit ÷ Revenue	39.6%	≥35%
Total Encounters	COUNTD(Encounter ID)	75,000	-
Average Wait Time	Average wait minutes	21.2 min	≤20 min
Patient Satisfaction	Average survey rating	4.52 / 5	≥4.5
Bed Utilization	Used capacity ÷ available capacity	77.8%	75–85%
Readmission Rate	Readmitted encounters ÷ eligible encounters	6.51%	≤6%
Cancellation Rate	Cancelled appointments ÷ appointments	6.0%	≤5%
Claim Collection Rate	Collected claim value ÷ claim value	Target benchmark shown at 90%	≥90%
Treatment Cost	SUM(Treatment Cost SAR)	SAR 66.5M	-
Avg Revenue / Encounter	Revenue ÷ encounters	SAR 1,468	-

Avg Cost / Encounter	Cost ÷ encounters	SAR 886	-
Avg Gross Profit / Encounter	Gross profit ÷ encounters	SAR 581	-
Survey Coverage	Distinct surveyed patients ÷ 30,000	20.0%	-
Revenue / Treatment Minute	Revenue ÷ treatment minutes	SAR 12.15	-

10. Dashboard Portfolio

01. Healthcare Executive Performance Overview

Executive-level summary of financial performance, encounter volume, patient satisfaction and trends.

KPI cards: Revenue; Gross Profit; Profit Margin; Total Encounters; Patient Satisfaction

Visuals: Monthly Revenue Trend; Monthly Encounter Trend; Revenue by Payer; Annual Revenue vs Cost

02. Patient & Utilization Analytics

Patient demand and utilization perspective designed to understand encounter mix, patient activity and service utilization.

03. Hospital Operations & Capacity

Facility and appointment operations view focused on capacity, utilization, cancellations, waiting time and departmental revenue.

KPI cards: Total Bed Capacity; Bed Utilization; Total Appointments; Cancellation Rate; Avg Wait Time

Visuals: Bed Utilization by Facility; Encounters by Facility; Appointment Status; Revenue by Department

04. Financial & Revenue Analytics

Financial performance view focused on cost, revenue efficiency, profitability and claims collection.

KPI cards: Total Treatment Cost; Avg Revenue per Encounter; Avg Cost per Encounter; Avg Gross Profit per Encounter; Outstanding Claim Value

Visuals: Monthly Revenue vs Cost; Gross Profit by Department; Revenue vs Cost by Service; Claim Collection Rate by Payer

05. Patient Experience & Quality

Quality and patient-experience view focused on survey feedback, outcomes and readmissions.

KPI cards: Survey Responses; Satisfaction Within Target; Readmission Rate; Positive Outcome Rate; Survey Coverage Rate

Visuals: Patient Satisfaction Rating Distribution; Patient Satisfaction Trend; Patient Outcomes Distribution; Readmission Rate by Encounter Type

06. Physician & Service Performance

Provider and service-level performance view focused on productivity, revenue contribution and treatment-time efficiency.

KPI cards: Total Physicians; Avg Encounters per Physician; Avg Revenue per Physician; Avg Service Time per Encounter; Revenue per Treatment Minute

Visuals: Physician Revenue Performance (Top 10); Encounters by Physician Specialty; Service Volume by Service (Top 10); Service Revenue Efficiency (Top 10)

11. Key Findings & Business Interpretation

Financial Performance

- Approximately SAR 110.1M in service revenue is generated against SAR 66.5M in treatment cost, producing approximately SAR 43.6M gross profit.
- Gross profit margin is 39.6%, above the illustrative 35% target.
- Average revenue per encounter is approximately SAR 1,468 versus SAR 886 average cost, producing approximately SAR 581 gross profit per encounter.
- Payer-level collection monitoring remains important because outstanding claims affect working capital.

Operations & Capacity

- Bed utilization is 77.8%, within the illustrative 75–85% target range.
- Average wait time is 21.2 minutes, slightly above the ≤ 20 -minute target.
- Appointment cancellation is 6.0%, above the $\leq 5\%$ target, indicating avoidable scheduling leakage.

Patient Experience & Quality

- Average patient satisfaction is 4.52/5, slightly above the 4.5 target.
- Readmission is 6.51%, above the $\leq 6\%$ target, making reduction a priority quality opportunity.
- Inpatient readmission is approximately 9.8%, higher than the outpatient and emergency rates shown in the dashboard.
- Survey coverage is 20%, so satisfaction results should be interpreted as survey-based rather than a complete view of all 30,000 patients.

Provider & Service Performance

- The network contains 60 physicians and averages 1,250 encounters per physician across the analytical period.
- Average revenue per physician is approximately SAR 1.8M; average service time per encounter is approximately 142 minutes.
- Revenue per treatment minute is approximately SAR 12.15, providing an efficiency lens for comparing services.

12. Recommendations

Priority	Recommended Action	Business Rationale
1. Reduce waiting time	Identify facility/department bottlenecks and manage performance against the 20-minute target.	Supports measurable improvement in operational, financial or quality performance.
2. Lower appointment cancellations	Analyze cancellation patterns and strengthen confirmations/follow-up for high-risk segments.	Supports measurable improvement in operational, financial or quality performance.
3. Prioritize readmission reduction	Focus clinical review on inpatient pathways, discharge planning, follow-up and high-readmission groups.	Supports measurable improvement in operational, financial or quality performance.
4. Protect the healthy utilization range	Maintain bed utilization around 75–85% while monitoring peak demand.	Supports measurable improvement in operational, financial or quality performance.
5. Strengthen claims collection	Prioritize weaker payer segments and monitor outstanding value alongside collection rate.	Supports measurable improvement in operational, financial or quality performance.
6. Scale high-efficiency services	Review services with strong revenue per treatment minute and strong volume for scalable practices.	Supports measurable improvement in operational, financial or quality performance.
7. Use physician performance for capacity planning	Combine volume, revenue and service-time measures to identify workload imbalance.	Supports measurable improvement in operational, financial or quality performance.
8. Increase survey coverage	Expand participation so patient-experience metrics represent a broader patient population.	Supports measurable improvement in operational, financial or quality performance.

13. Technical Implementation

Tools & Technologies

Item	Details
Python	Data preparation and transformation
Pandas	Tabular manipulation and validation
Google Colab	Notebook execution
Microsoft Excel	Structured analytical workbook storage
Tableau Public	Relationship modeling, calculations, visualization and dashboards

Analytical Design Principles

- Use descriptive dimensions for consistent filtering.

- Keep independent facts at natural grain.
- Use relationships rather than flattening unrelated facts.
- Use COUNTD for entity-level KPIs where appropriate.
- Use target reference lines to make performance actionable.
- Use Top N views for long category lists.
- Separate dashboards by decision domain.

Representative Calculations

Measure	Logic
Gross Profit	Service Revenue SAR – Treatment Cost SAR
Profit Margin	Gross Profit ÷ Service Revenue SAR
Avg Revenue / Encounter	Service Revenue SAR ÷ COUNTD(Encounter ID)
Avg Cost / Encounter	Treatment Cost SAR ÷ COUNTD(Encounter ID)
Avg Gross Profit / Encounter	Gross Profit ÷ COUNTD(Encounter ID)
Avg Encounters / Physician	COUNTD(Encounter ID) ÷ COUNTD(Physician ID)
Revenue / Treatment Minute	Service Revenue SAR ÷ Treatment Time Minutes
Survey Coverage Rate	COUNTD(Patient ID in PATIENT_SURVEYS) ÷ 30,000

14. Validation & Quality Assurance

Quality assurance was applied at both the Python export stage and the Tableau modeling/dashboard stage.

Data-Level QA

- Confirmed expected row and column counts for all dimension and fact tables.
- Validated revenue, cost and gross profit control totals.
- Checked average wait time, satisfaction and readmission control results.
- Confirmed date coverage across 2023–2025.
- Confirmed required keys for Tableau relationships.

Tableau-Level QA

- Built relationships at the correct key level.
- Used multiple base facts rather than physically joining unrelated facts.
- Reviewed KPI values against known control totals.
- Applied target/reference lines for operational and quality benchmarks.
- Reviewed dashboard layout, label visibility, chart readability and KPI presentation.
- Exported all six completed dashboards as PDFs.

Core Control Totals

Item	Details
Service Revenue	SAR 110,077,066.91
Treatment Cost	SAR 66,470,103.04

Gross Profit	SAR 43,606,963.87
Average Wait Time	21.2 minutes
Average Satisfaction	4.52 / 5
Readmission Rate	6.51%

15. Project Deliverables & Folder Structure

The project is organized to separate source data, transformed analytical data, reproducible preparation work, Tableau assets and dashboard outputs.

Folder	Purpose
01_Original_Dataset	Original Excel dataset used as the starting point.
02_Transformed_Dataset	Validated final analytical workbook prepared for Tableau.
03_Notebook	Google Colab/Jupyter notebook containing the preparation workflow.
04_Tableau	Tableau workbook contains relationship models and dashboards.
05_Dashboards	Six PDF dashboard exports.
06_Documentation	README, project documentation and supporting material.

16. Limitations & Assumptions

- The dataset is synthetic and should not be interpreted as real patient or hospital information.
- Clinical, financial and operational patterns are designed for portfolio demonstration rather than production decision-making.
- Illustrative KPI targets are management benchmarks for demonstration and would require formal definition in a real organization.
- Survey coverage is 20%, so patient-experience measures may be subject to response bias.
- Readmission and outcome measures depend on definitions and flags embedded in the synthetic data.
- Production deployment would require governance, privacy controls, data-quality monitoring, access management and formal metric definitions.

17. Conclusion

Healthcare Operations & Patient Performance Analytics demonstrates a complete Business Intelligence workflow from raw healthcare data through Python-based preparation, relational modeling, KPI design, Tableau visualization and portfolio-ready reporting.

The solution provides a structured view of the trade-offs between financial performance, patient demand, capacity, operational efficiency, quality and provider productivity. The multi-fact Tableau architecture is a key technical feature because it enables cross-domain analysis without compromising the grain of individual healthcare processes.

The completed six-dashboard portfolio turns a complex synthetic healthcare dataset into a management-oriented analytical product. The project is ready for final Tableau publishing, GitHub packaging, README development and professional portfolio presentation.

Appendix A - Dashboard KPI Reference

Dashboard	Primary KPIs	Decision Focus
Executive Performance Overview	Revenue; Gross Profit; Profit Margin; Encounters; Satisfaction	Enterprise performance
Patient & Utilization Analytics	Patient and utilization measures	Demand and utilization
Hospital Operations & Capacity	Bed Capacity; Bed Utilization; Appointments; Cancellation; Wait Time	Capacity and operations
Financial & Revenue Analytics	Treatment Cost; Avg Revenue/Cost/Profit per Encounter; Outstanding Claims	Financial efficiency
Patient Experience & Quality	Survey Responses; Satisfaction; Readmission; Outcomes; Survey Coverage	Quality and patient experience
Physician & Service Performance	Physicians; Encounters/Physician; Revenue/Physician; Service Time; Revenue/Minute	Provider and service productivity

Appendix B - Project Naming Standard

The project uses the standardized project name “Healthcare Operations & Patient Performance Analytics” across documentation, dashboard assets and portfolio references. File names use numbered prefixes for ordered dashboard presentation and underscores for portability across Windows, GitHub and cloud storage.